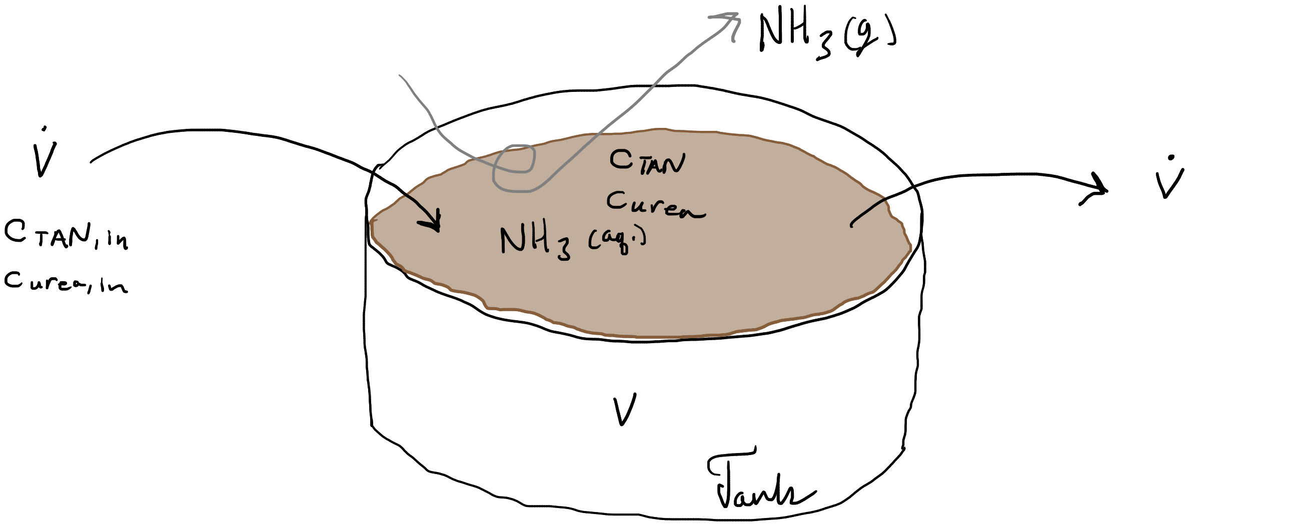
More details on the conceptual and mathematical models for the ammonia emission problem

A tank open to the air receives manure at a constant volumetric flow rate and it is removed at the same constant rate. The fresh manure has a fixed urea and total ammoniacal nitrogen (TAN) concentration. Urea can hydrolyze to ammonia (although it does it rather quickly without an inhibitor) and ammonia can volatilize from the tank (although only the uncharged NH3 species is volatile).



$\dot{V}$ = constant manure flow in and out $\quad m^3\,s^{-1}$

V = constant manure volume in tank $\quad m^3$

C_{TAN} = total ammoniacal N concentration $\quad kg\,m^{-3}$

C_{urea} = urea concentration as N $\quad kg\,m^{-3}$

mass balance:

TAN accumulation = TAN pumping in - NH3 volatilization out + TAN generation - TAN pumping out

Urea accumulation = Urea pumping in - urea hydrolysis - urea pumping out

The dynmod() function uses a mass balance normalized by manure volume V. So units are kg/m3-s (kg of N) for each term.

23 Oct 2025. Correction to NH3 volatilization below (was * A now / d becaue *A / V = /d.

The constitutive equations are given below. These also have units of kg/m3-s because of the normalization by manure volume. This is also by we have retention time and not flow below.

TAN pumping in = $C_{TAN,in} / \tau$ — Retention time = $\frac{V}{\dot{V}}$ (τ is "tau") (s)

TAN pumping out = C_{TAN} / τ — Tank conc.

NH3 volatilization out = $k_L / d \cdot (f_{NH_3} \cdot C_{TAN} - C_{NH_3,bg} / H)$

gas-phase conc. ($kg\,m^{-3}$)

TAN generation = $\alpha \cdot C_{urea}$

Henry's law constant aq:g

Urea pumping in = $C_{urea,in} / \tau$

Urea hydrolysis = $\alpha \cdot C_{urea}$

Urea pumping out = C_{urea} / τ — First-order rate constant (s^{-1})